

ASHUTOSH SHARMA

Generative AI / LLM Engineer

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SUMMARY

Generative AI Engineer with 3.5 years at T-Systems building production RAG and agentic-LLM systems for enterprise document intelligence. Works across the full LLM stack on LlamaIndex, LangGraph, FastAPI, and pgvector: hybrid retrieval, multi-tool agents, RAGAS/Phoenix evaluation, and sub-second async streaming. Designed a gVisor-sandboxed agent runtime mapped to the OWASP LLM Top 10, on an applied-ML and analytics background. Open-source author of [pi-coding-agent](#), published on PyPI.

TECHNICAL SKILLS

Generative AI & LLMs	RAG (hybrid search, reranking, CRAG), knowledge-graph RAG (GraphRAG), agentic AI, multi-agent systems, LLM agents & ReAct, tool/function calling, MCP (Model Context Protocol), structured generation, prompt engineering, embeddings, vector & semantic search, guardrails, LLM evaluation (RAGAS), LLM observability (Phoenix)
Frameworks	LlamaIndex, LangGraph, LangChain, CrewAI, Hugging Face Transformers
Models	Llama 3, GPT, Claude
Backend & Serving	Python (async), FastAPI, PostgreSQL + pgvector, SQLAlchemy (async), Celery, RabbitMQ, WebSocket/SSE streaming, REST APIs, SQL, Pydantic
Cloud & MLOps	AWS, Docker, Kubernetes, Helm, CI/CD, Git, object storage (S3), Prometheus, Loguru, MLOps
Machine Learning	PyTorch, scikit-learn, deep learning, NLP, classification, topic modeling (LDA), XGBoost, LightGBM

PROFESSIONAL EXPERIENCE

AI Engineer · T-Systems ICT India Pvt. Ltd.

Dec 2022 – Present

AI Operator: Agentic RAG & Sandboxed Agent Platform (May 2025 – Present)

- Built and own the backend of an agentic RAG platform on FastAPI + LlamaIndex answering questions over 10,000+ enterprise documents; hybrid retrieval (semantic ANN + BM25) on a custom pgvector store with Cohere reranking and streamed generation through an internal LLM gateway reduced p95 answer latency about 35% under production load.
- Designed a gVisor-sandboxed agent runtime on Kubernetes: per-session isolated pods run LangGraph/deepagents in a two-container split that keeps secrets out of the model's code workspace, so generated code never touches host or network.
- Partnered with the security team on a 12-layer defense-in-depth model (syscall filtering, network-egress control, path sandboxing, per-agent quotas, secret scrubbing) mapped to the OWASP LLM Top 10, treating the LLM as an untrusted actor.
- Engineered six-tool agentic workflows on LlamaIndex's Workflow API (query decomposition, parallel ReAct search, synthesis, citation/source filtering) and raised RAGAS faithfulness from roughly 40% to over 80% via reranking and eval-driven prompt tuning, with loop guards on the ReAct agent.
- Stood up a RAGAS + Phoenix evaluation harness running 1,000+ automated evals a week to tune prompts and retrieval parameters, and presented findings to engineering leadership.

- Scaled ingestion from PDF-only to six formats across 15+ enterprise sources on Celery/RabbitMQ workers with per-child memory caps, and served async SSE streaming at sub-second time-to-first-token behind Keycloak JWT auth with per-tenant isolation.
- Shipped a document-to-story feature that parses uploaded PDFs/DOCX in memory and generates agile Epics and user stories over streaming chat, including Epic-level Mermaid diagrams.

Insurance Healthcare Agent: Agentic RAG *(Dec 2024 – Apr 2025)*

- Developed an agentic policy assistant combining document retrieval with structured PostgreSQL lookups via function-calling tools and low-confidence fallback handling; trimmed query-resolution time about 40% and average wait time about 70%.
- Ran it in production at 10,000+ queries a month with 99%+ uptime and graceful degradation under load.

CrewAI Multi-Agent Content System *(Jul 2024 – Nov 2024)*

- Created a CrewAI multi-agent content pipeline with dedicated researcher, writer, and editor agents handing off through a shared workflow across five content types, halving draft turnaround.

Topic Modeling for Document Clustering *(Feb 2024 – Jul 2024)*

- Clustered 1,000+ enterprise documents with LDA topic modeling, then trained a LightGBM classifier (about 90% accuracy) on the topics to automate categorization that had been manual.

Enterprise Data Analytics & ML *(Dec 2022 – Jan 2024)*

- Trained customer-segmentation and classification models (XGBoost, Random Forest, ~87% accuracy) and assembled the ETL and feature-engineering pipelines feeding them over 50K+ records, plus analytics dashboards for enterprise clients.

Intern Data Scientist · Analytics Labs

Nov 2021 – Nov 2022

- Lowered targeted-segment churn about 12% with a Random Forest attrition model, and defined real-time KPIs across 500+ devices.

OPEN SOURCE

pi-coding-agent (open source · PyPI, v0.6.0) — Authored a transparent, multi-provider AI coding agent (Claude / GPT / Gemini / Ollama) with a sandboxed executor, planner, sub-agents, a skills system, and MCP tool support.

EDUCATION

B.Sc. Computer Science, Mathematics & Statistics · Bangalore North University 2018 – 2021

M.Sc. Data Science · Manipal Academy of Higher Education 2022 – 2024

CERTIFICATIONS

Generative AI with Large Language Models — DeepLearning.AI / Coursera · LangChain for LLM Application Development — DeepLearning.AI · Machine Learning Specialization — Coursera (Andrew Ng)

LANGUAGES

English (professional working) · Hindi (native)